

SET UP ROUTING MODEL

GLY606 Water Data Analysis & Modeling

Nov 15th 2024



Directly go to
<https://jupyterhub.cuahsi.org/>

Let's use “**WRF-Hydro
v5.1 – NWM v2.0**”
environment

Server Options

- **Python v3.9.7 - JupyterLab Interface**

This is a general-purpose scientific computing environment built on Python version 3.9.7. When using this server, you will be working through the JupyterLab interface. It comes with pre-configured, widely used scientific libraries including scipy, pandas, geopandas, sympy, scikit-learn, matplotlib, ulmo, landlab, gdal, xarray, TauDEM, hscient, dataretrieval, rasterio, pyproj, fiona, geemap, fsspec, and others.

- **R v4.2.3 - JupyterLab Interface**

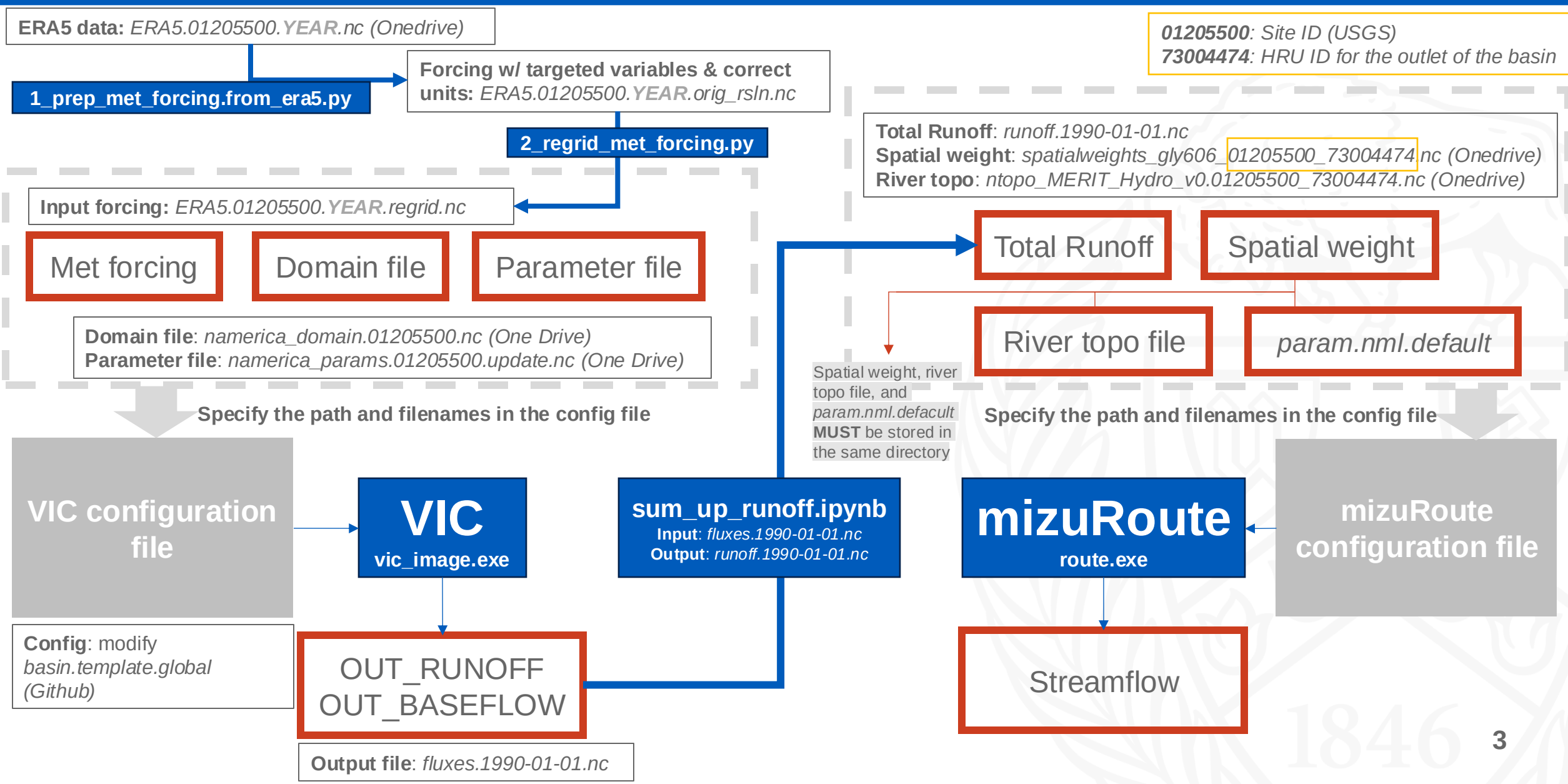
This serves as a general-purpose scientific computing environment built on the R programming language. When using this server, you will have access to both the JupyterLab and RStudio interfaces. The environment is pre-configured with commonly used tools and libraries, including randomForest, forecast, sf, rjsonio, ncd4, ggmap, dataRetrieval, and WaterML.

- **SUMMA Modeling**

This is a computing environment built on Python version 3.9 and pre-installed tools to support using the Structure for Unifying Multiple Modeling Alternative (SUMMA). SUMMA is a hydrologic modeling framework that can be used to configure a wide range of hydrological model analysis. When using this server, you will be working through the JupyterLab interface. The environment comes with limited pre-installed scientific Python packages. For more information on the SUMMA model, visit the project [webpage](#).

- **WRF-Hydro v5.1 - NWM v2.0**

This computing environment is structured around Python version 3.9 and is tailored for seamless operation with the WRF-Hydro model because it has the source code pre-compiled. WRF-Hydro is a community hydrological model suitable for a range of projects, including real-time flash flood prediction, long-term water supply projections, and evaluations of regional hydroclimate impacts. Notably, WRF-Hydro has been used as the core hydrological module within the National Water Model (NWM) versions 2.0 and 2.1. When using this server, you will be working through the JupyterLab interface. While the environment features a curated selection of pre-installed scientific Python packages, users can follow CUAHSI's guideline to create a customized conda environment, and then install specific Python packages that cater to their unique project requirements. For more information on the WRF-Hydro model, visit the project [website](#).



Step 1: Calculate total runoff

- Total runoff = OUT_RUNOFF (surface runoff) + OUT_BASEFLOW (subsurface runoff)
- Please use "sum_up_runoff.ipynb"
 - Please remember to use the virtual environment - [xesmf_env](#)
 - Please copy this notebook to your VIC output directory ([~/model/final_project/vic/output](#))
 - This notebook is located in Github: [GLY606_2024/final_project/vic_postprocess](#)
 - Remember to update the input and output filenames

Step 2: Create a model run folder

- **cd** ~/model/final_project/mizuroute
- **mkdir** ancillary_data input output settings

Example folder structure

- model
 - final_project
 - vic
 - mizuroute
 - ancillary_data
 - input
 - output
 - settings

Step 3: Move files to the targeted folders

- `~/model/final_project/mizuroute/ancillary_data`
 - `spatialweights_gly606_<siteid>_<outlet_id>.nc`
 - `ntopo_MERIT_Hydro_v0.<siteid>_<outlet_id>.nc`
 - `param.nml.default`
- `~/model/final_project/vic/settings`
 - `route.control.template`

We need to replace the entire string including the bracket with the targeted site ID and outlet ID. For example, for the basin with site ID 01205500 and outlet ID 73004474, the spatial weight file would be

`spatialweights_gly606_01205500_73004474.nc`

`route.control.template` is located in Github: [GLY606_2024/final_project/mizuroute_config_template](https://github.com/GLY606_2024/final_project/mizuroute_config_template)

Step 4: Prepare configuration file

- `cd ~/model/final_project/mizuroute/settings`
- `cp route.control.template route.control.<siteid>`
- `vim route.control.<siteid>`

- <STARTYEAR>	1990	Example input values
- <ENDYEAR>	1990	
- <SITEID>	01205500	
- <RUNOFF_FILENAME>	runoff.1990-01-01.nc	
- <NTOPO_FILENAME>	ntopo_MERIT_Hydro_v0.01205500_73004474.nc	
- <SPATIAL_WEIGHT_FILENAME>	spatialweights_gly606_01205500_73004474.nc	

If you followed the folder structure as I suggested, you do not need to modify the path to the files above.

```
! *****
! DEFINE DIRECTORIES
! -----
<ancil_dir>      ../ancillary_data/
<input_dir>     ../../vic/output/
<output_dir>    ../output/
! *****
```

Otherwise, you will need to modify the path to the directories

Type **:wq** to save and exit the vim editor

Step 5: Run the model

- `cd ~/model/final_project/mizuroute/settings`
- `~/model/mizuRoute/route/bin/route.exe route.control.01205500 > run.log`

Replace this using
your own site ID

Step 6: Take a look at the output

- Example notebook for Homework 8
 - https://github.com/act-hydro/GLY606_2024/tree/main/final_project/compare_model_output
 - Important to note
 - Unit conversion
 - The unit for the USGS streamflow is **ft³/s**
 - The unit for the mizuRoute streamflow is **m³/s**
 - Variable name for mizuRoute streamflow is **KWroutedRunoff**